

Dhairya Shah Lecture 19 Scribe



L19.pdf
PDF

I want you to generate a Lecture Scribe that can be used for Exam Preparation. Make sure to use only the lecture slides shared with you. List of things that scribe should have are: Definitions and notation, Assumptions and conditions, Statements of theorems or results, Step by step proofs, Worked examples with all intermediate steps included. Everything should be from the lecture slides. Include step by step reasoning within the final scribe for maximum clarity for revision. Do not introduce any external intuition, creative explanations, or examples not found in the provided materials. Remember that this lecture scribe is for revision purpose of exam. Important: I will use the same prompt and the same context materials for multiple independent runs to generate different outputs. After that I will select the best scribe that is been given to me.

Here is a **Lecture Scribe for Exam Preparation (Lecture 19: N-Variate Joint PDF/PMF/CDF)** strictly constructed from the provided slides .

LECTURE SCRIBE — CSE400 Lecture 19

N-Variate Joint PDF / PMF / CDF

1. DEFINITIONS AND NOTATION

1.1 Joint Representation of Random Variables

For random variables $X_1, X_2, \dots, X_N$, the vector form is:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

- The **joint distribution** describes the probability behavior of all variables together.
 - It can be represented using:
 - Joint PMF
 - Joint PDF
 - Joint CDF
- (Ref: Page 17)

1.2 Joint PMF (Discrete Case)

$$P_{X_1, X_2, \dots, X_N}(x_1, x_2, \dots, x_N) = P(X_1 = x_1, X_2 = x_2, \dots, X_N = x_N)$$

(Ref: Page 22)

1.3 Joint CDF

$$F_{X_1, X_2, \dots, X_N}(x_1, x_2, \dots, x_N) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_N \leq x_N)$$

(Ref: Page 23)

1.4 Joint PDF (Continuous Case)

$$f_{X_1,X_2,\dots,X_N}(x_1,x_2,\dots,x_N) = \frac{\partial^N}{\partial x_1 \partial x_2 \dots \partial x_N} F_{X_1,X_2,\dots,X_N}(x_1,x_2,\dots,x_N)$$

(Ref: Page 24)

1.5 Independence of Random Variables

Let:

$$\mathbf{X} = [X_1, X_2, \dots, X_N]^T$$

If independent:

$$f_X(x_1,x_2,\dots,x_N) = \prod_{i=1}^N f_{X_i}(x_i)$$

(Ref: Page 25)

2. JOINT GAUSSIAN DISTRIBUTION

2.1 Joint Gaussian PDF (Two Variables)

For X, Y :

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(\frac{-1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) \right]\right)$$

(Ref: Page 3, 5)

Parameters

- μ_X, μ_Y : means
- σ_X, σ_Y : standard deviations
- ρ : correlation coefficient

3. EXAMPLE 1: JOINT GAUSSIAN VARIABLES

3.1 Given

- $\mu_X = 2, \sigma_X^2 = 1 \Rightarrow \sigma_X = 1$
- $\mu_Y = -3, \sigma_Y^2 = 4 \Rightarrow \sigma_Y = 2$
- $\text{cov}(X,Y) = -1$

(Ref: Page 4)

3.2 Step-by-Step Solution (a): Joint PDF

Step 1: Correlation Coefficient

$$\rho_{XY} = \frac{\text{cov}(X,Y)}{\sigma_X\sigma_Y} = \frac{-1}{(1)(2)} = -\frac{1}{2}$$

(Ref: Page 7–8)

Step 2: Substitute into General Formula

$$\mu_X = 2, \sigma_X = 1, \mu_Y = -3, \sigma_Y = 2, \rho = -0.5$$

(Ref: Page 9)

Step 3: Final Joint PDF

$$f_{XY}(x,y) = \frac{1}{2\pi\sqrt{3}} \exp\left(-\frac{2}{3}\left[(x-2)^2 + \frac{(x-2)(y+3)}{2} + \frac{(y+3)^2}{4}\right]\right)$$

(Ref: Page 10)

3.3 Solution (b): Marginal PDFs

Definition

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y) dy \quad f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y) dx$$

(Ref: Page 11)

Key Property

If jointly Gaussian:

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

(Ref: Page 11)

Final Marginals

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-2)^2}{2}\right)$$
$$f_Y(y) = \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{(y+3)^2}{8}\right)$$

(Ref: Page 12)

3.4 Solution (c): Probability $P(X < 0)$

Step 1: Identify CDF

$$F_X(x) = P(X \leq x) \Rightarrow F_X(0) = P(X < 0)$$

(Ref: Page 13)

Step 2: Standardization

$$Z = \frac{X - \mu_X}{\sigma_X}$$
$$P(X < 0) = P\left(Z < \frac{0-2}{1}\right) = P(Z < -2) = \Phi(-2)$$

(Ref: Page 14)

Step 3: Symmetry

$$\Phi(-x) = 1 - \Phi(x) = P(Z > x)$$
$$P(Z < -2) = P(Z > 2)$$

(Ref: Page 15)

Step 4: Q-function

$$Q(x) = P(Z > x)$$

Final Answer

$$P(X < 0) = Q(2)$$

(Ref: Page 16)

4. APPLICATION: LARGE MIMO SYSTEM

System Setup

- M : Transmit antennas
- N : Receive antennas

(Ref: Page 18–19)

Channel Model

- Each path modeled as random variable h_{ij}

Channel Matrix

$$H = \begin{bmatrix} h_{11} & h_{12} & \dots & h_{1N} \\ h_{21} & h_{22} & \dots & h_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ h_{M1} & h_{M2} & \dots & h_{MN} \end{bmatrix}$$

(Ref: Page 21)

5. EXAMPLE 2: UNIFORMLY DISTRIBUTED 3D VECTOR

5.1 Given

$$|x_1| \leq 1, \quad |x_2| \leq 1, \quad |x_3| \leq 1$$

$$f_{X_1,X_2,X_3}(x_1,x_2,x_3) = \begin{cases} C & \text{inside cube} \\ 0 & \text{otherwise} \end{cases}$$

(Ref: Page 26)

5.2 (a) Find Constant C

$$\int_{-1}^1 \int_{-1}^1 \int_{-1}^1 C \, dx_1 dx_2 dx_3 = 1$$

$$C(2)(2)(2) = 1 \Rightarrow C = \frac{1}{8}$$

(Ref: Page 28)

5.3 (b) Marginal $f_{X_1,X_2}(x_1,x_2)$

$$f_{X_1,X_2}(x_1,x_2)=\int_{-1}^1\frac{1}{8}dx_3=\frac{1}{8}(2)=\frac{1}{4}$$

(Ref: Page 29)

5.4 (c) Marginal $f_{X_1}(x_1)$

$$f_{X_1}(x_1)=\int_{-1}^1\int_{-1}^1\frac{1}{8}dx_2dx_3=\frac{1}{2}$$

(Ref: Page 31)

5.5 (d) Conditional PDF

$$f_{X_1|X_2,X_3}(x_1|x_2,x_3)=\frac{f_{X_1,X_2,X_3}(x_1,x_2,x_3)}{f_{X_2,X_3}(x_2,x_3)}=\frac{1/8}{1/4}=\frac{1}{2}$$

(Ref: Page 33)

5.6 (e) Conditional PDF

$$f_{X_1,X_2|X_3}(x_1,x_2|x_3)=\frac{f_{X_1,X_2,X_3}}{f_{X_3}}=\frac{1/8}{1/2}=\frac{1}{4}$$

(Ref: Page 35)

5.7 (f) Independence Check

LHS

$$f_{X_1,X_2,X_3}=\frac{1}{8}$$

RHS

$$f_{X_1}f_{X_2}f_{X_3}=\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2}=\frac{1}{8}$$

Conclusion

$$\text{LHS} = \text{RHS} \Rightarrow \text{Independent}$$

(Ref: Page 37–39)

6. EXAMPLE 3: TRIVARIATE DISTRIBUTION

6.1 Given

$$f_{XYZ}(x,y,z)=\begin{cases} Ke^{-(ax+by+cz)} & x,y,z>0 \\ 0 & \text{otherwise} \end{cases}$$

(Ref: Page 40)

6.2 (a) Find K

$$\int_0^\infty\int_0^\infty\int_0^\infty Ke^{-(ax+by+cz)}dx\,dy\,dz=1$$

Factorization:

$$K \left(\int_0^\infty e^{-ax} dx \right) \left(\int_0^\infty e^{-by} dy \right) \left(\int_0^\infty e^{-cz} dz \right) = 1$$

$$K \left(\frac{1}{a} \right) \left(\frac{1}{b} \right) \left(\frac{1}{c} \right) = 1$$

$$K = abc$$

(Ref: Page 42)

6.3 (b) Marginal Joint PDF $f_{XY}(x, y)$

$$f_{XY}(x, y) = \int_0^\infty f_{XYZ}(x, y, z) dz = \int_0^\infty abce^{-(ax+by+cz)} dz$$

$$= abe^{-(ax+by)} \left[\int_0^\infty ce^{-cz} dz \right] = abe^{-(ax+by)}$$

(Ref: Page 44)

6.4 (c) Marginal PDF $f_X(x)$

$$f_X(x) = \int_0^\infty f_{XY}(x, y) dy = \int_0^\infty abe^{-(ax+by)} dy$$

$$= ae^{-ax}$$

(Ref: Page 46)

6.5 (d) Independence Check**LHS**

$$f_{XYZ}(x, y, z) = abce^{-(ax+by+cz)}$$

RHS

$$f_X(x)f_Y(y)f_Z(z) = (ae^{-ax})(be^{-by})(ce^{-cz}) = abce^{-(ax+by+cz)}$$

Conclusion

$$\text{LHS} = \text{RHS} \Rightarrow X, Y, Z \text{ are independent}$$

(Ref: Page 48–50)

END OF SCRIBE

This scribe strictly follows the lecture slides, preserving:

- Definitions and notation
- Conditions and assumptions
- Exact formulas
- Step-by-step derivations
- Fully worked examples with intermediate steps

No external material has been introduced.



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Here is an **alternative Lecture Scribe (Exam-Oriented, Fully Stepwise, Strictly from Slides)** based only on the provided material .

LECTURE SCRIBE — CSE400 (Lecture 19)

N-Variate Joint PDF / PMF / CDF

1. DEFINITIONS AND NOTATION

1.1 Vector Representation of Random Variables

For N random variables:

$$\mathbf{X} = [X_1, X_2, \dots, X_N]^T$$

- The **joint distribution** captures the behavior of all variables together.
- Representation forms:
 - Joint PMF
 - Joint PDF
 - Joint CDF

(Ref: Page 17)

1.2 Joint PMF

For discrete random variables:

$$P_{X_1, \dots, X_N}(x_1, \dots, x_N) = P(X_1 = x_1, \dots, X_N = x_N)$$

(Ref: Page 22)

1.3 Joint CDF

$$F_{X_1, \dots, X_N}(x_1, \dots, x_N) = P(X_1 \leq x_1, \dots, X_N \leq x_N)$$

(Ref: Page 23)

1.4 Joint PDF

$$f_{X_1, \dots, X_N}(x_1, \dots, x_N) = \frac{\partial^N}{\partial x_1 \cdots \partial x_N} F_{X_1, \dots, X_N}(x_1, \dots, x_N)$$

(Ref: Page 24)

1.5 Independence Condition

If $X_1, \dots, X_N$ are independent:

$$f_{X_1, \dots, X_N}(x_1, \dots, x_N) = \prod_{i=1}^N f_{X_i}(x_i)$$

(Ref: Page 25)

2. JOINT GAUSSIAN RANDOM VARIABLES

2.1 Statement (Joint Gaussian PDF)

For two variables X, Y :

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(\frac{-1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) \right]\right)$$

(Ref: Page 3, 5)

2.2 Notation

- μ_X, μ_Y : means
- σ_X, σ_Y : standard deviations
- ρ : correlation coefficient

3. EXAMPLE 1 — JOINT GAUSSIAN

3.1 Given Conditions

- $\mu_X = 2, \sigma_X^2 = 1$
- $\mu_Y = -3, \sigma_Y^2 = 4$
- $\text{cov}(X, Y) = -1$

(Ref: Page 4)

3.2 (a) Joint PDF Construction

Step 1: Compute Standard Deviations

$$\sigma_X = \sqrt{1} = 1, \quad \sigma_Y = \sqrt{4} = 2$$

(Ref: Page 8)

Step 2: Compute Correlation

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X\sigma_Y} = \frac{-1}{(1)(2)} = -\frac{1}{2}$$

(Ref: Page 7–8)

Step 3: Substitute into General Formula

Values:

$$\mu_X = 2, \sigma_X = 1, \mu_Y = -3, \sigma_Y = 2, \rho = -0.5$$

(Ref: Page 9)

Step 4: Final Expression

$$f_{XY}(x,y) = \frac{1}{2\pi\sqrt{3}} \exp\left(-\frac{2}{3}\left[(x-2)^2 + \frac{(x-2)(y+3)}{2} + \frac{(y+3)^2}{4}\right]\right)$$

(Ref: Page 10)

3.3 (b) Marginal PDFs

Step 1: Definition

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y) dy \quad f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y) dx$$

(Ref: Page 11)

Step 2: Property

If jointly Gaussian:

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

(Ref: Page 11)

Step 3: Final Marginals

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-2)^2}{2}}$$
$$f_Y(y) = \frac{1}{2\sqrt{2\pi}} e^{-\frac{(y+3)^2}{8}}$$

(Ref: Page 12)

3.4 (c) Compute $P(X < 0)$

Step 1: Use CDF

$$F_X(x) = P(X \leq x) \Rightarrow P(X < 0) = F_X(0)$$

(Ref: Page 13)

Step 2: Standardization

$$Z = \frac{X - \mu_X}{\sigma_X}$$
$$P(X < 0) = P\left(Z < \frac{0-2}{1}\right) = P(Z < -2)$$

(Ref: Page 14)

Step 3: Symmetry

$$\Phi(-x) = 1 - \Phi(x) = P(Z > x)$$
$$P(Z < -2) = P(Z > 2)$$

(Ref: Page 15)

Step 4: Q-function Definition

$$Q(x) = P(Z > x)$$

Final Result

$$P(X < 0) = Q(2)$$

(Ref: Page 16)

4. APPLICATION: LARGE MIMO SYSTEM

4.1 Assumptions

- M transmit antennas
- N receive antennas

(Ref: Page 18–19)

4.2 Channel Model

Each link:

h_{ij} is a random variable

(Ref: Page 20)

4.3 Channel Matrix

$$H = \begin{bmatrix} h_{11} & \cdots & h_{1N} \\ h_{21} & \cdots & h_{2N} \\ \vdots & \ddots & \vdots \\ h_{M1} & \cdots & h_{MN} \end{bmatrix}$$

(Ref: Page 21)

5. EXAMPLE 2 — UNIFORM 3D VECTOR

5.1 Given

$$|x_1| \leq 1, \quad |x_2| \leq 1, \quad |x_3| \leq 1$$

$$f_{X_1,X_2,X_3} = \begin{cases} C & \text{inside cube} \\ 0 & \text{otherwise} \end{cases}$$

(Ref: Page 26)

5.2 (a) Find C

Step 1: Total Probability = 1

$$\int_{-1}^1 \int_{-1}^1 \int_{-1}^1 C \, dx_1 dx_2 dx_3 = 1$$

Step 2: Evaluate

$$C(2)(2)(2) = 1 \Rightarrow C = \frac{1}{8}$$

(Ref: Page 28)

5.3 (b) Marginal f_{X_1,X_2}

$$f_{X_1,X_2} = \int_{-1}^1 \frac{1}{8} dx_3 = \frac{1}{8}(2) = \frac{1}{4}$$

(Ref: Page 29)

5.4 (c) Marginal f_{X_1}

$$f_{X_1} = \int_{-1}^1 \int_{-1}^1 \frac{1}{8} dx_2 dx_3$$

Step-by-step:

$$= \int_{-1}^1 \left(\frac{1}{8} \cdot 2 \right) dx_2 = \int_{-1}^1 \frac{1}{4} dx_2 = \frac{1}{4}(2) = \frac{1}{2}$$

(Ref: Page 31)

5.5 (d) Conditional PDF

$$f_{X_1|X_2,X_3} = \frac{f_{X_1,X_2,X_3}}{f_{X_2,X_3}} = \frac{1/8}{1/4} = \frac{1}{2}$$

(Ref: Page 33)

5.6 (e) Conditional PDF

$$f_{X_1,X_2|X_3} = \frac{f_{X_1,X_2,X_3}}{f_{X_3}} = \frac{1/8}{1/2} = \frac{1}{4}$$

(Ref: Page 35)

5.7 (f) Independence Proof

Condition

$$f_{X_1,X_2,X_3} = f_{X_1} f_{X_2} f_{X_3}$$

Step 1: LHS

$$= \frac{1}{8}$$

Step 2: RHS

$$= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

Conclusion

$$\text{LHS} = \text{RHS} \Rightarrow \text{Independent}$$

(Ref: Page 37–39)

6. EXAMPLE 3 — TRIVARIATE EXPONENTIAL

6.1 Given

$$f_{XYZ}(x, y, z) = \begin{cases} K e^{-(ax+by+cz)} & x, y, z > 0 \\ 0 & \text{otherwise} \end{cases}$$

(Ref: Page 40)

6.2 (a) Find K

Step 1: Total Probability

$$\int_0^\infty \int_0^\infty \int_0^\infty K e^{-(ax+by+cz)} dx dy dz = 1$$

Step 2: Factorization

$$K \left(\int_0^\infty e^{-ax} dx \right) \left(\int_0^\infty e^{-by} dy \right) \left(\int_0^\infty e^{-cz} dz \right) = 1$$

Step 3: Evaluate Integrals

$$= \frac{1}{a} \cdot \frac{1}{b} \cdot \frac{1}{c}$$

Step 4: Solve

$$K \frac{1}{abc} = 1 \Rightarrow K = abc$$

(Ref: Page 42)

6.3 (b) Marginal $f_{XY}(x, y)$

$$f_{XY}(x, y) = \int_0^\infty abce^{-(ax+by+cz)} dz$$

Step-by-step:

$$\begin{aligned} &= abe^{-(ax+by)} \int_0^\infty ce^{-cz} dz \\ &= abe^{-(ax+by)} \end{aligned}$$

(Ref: Page 44)

6.4 (c) Marginal $f_X(x)$

$$\begin{aligned} f_X(x) &= \int_0^\infty abe^{-(ax+by)} dy \\ &= ae^{-ax} \end{aligned}$$

(Ref: Page 46)

6.5 (d) Independence Proof

Step 1: LHS

$$f_{XYZ} = abce^{-(ax+by+cz)}$$

Step 2: RHS

$$f_X f_Y f_Z = (ae^{-ax})(be^{-by})(ce^{-cz}) = abce^{-(ax+by+cz)}$$

Conclusion

$$\text{LHS} = \text{RHS} \Rightarrow \text{Independent}$$

(Ref: Page 48–50)

END OF LECTURE SCRIBE

This version emphasizes:

- Clear separation of assumptions
- Step-by-step derivations exactly as shown in slides
- Explicit intermediate steps in integrations and substitutions
- No additional explanations beyond slide content



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LECTURE SCRIBE — CSE400 Lecture 19

N-Variate Joint PDF / PMF / CDF

1. DEFINITIONS, NOTATION, AND REPRESENTATION

1.1 Vector Form of Random Variables

For N random variables:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

- Joint distribution describes the probability behavior of all variables together.
- Representations:
 - Joint PMF
 - Joint PDF
 - Joint CDF

(Ref: Page 17)

1.2 Joint PMF (Discrete Variables)

$$P_{X_1, \dots, X_N}(x_1, \dots, x_N) = P(X_1 = x_1, \dots, X_N = x_N)$$

(Ref: Page 22)

1.3 Joint CDF

$$F_{X_1, \dots, X_N}(x_1, \dots, x_N) = P(X_1 \leq x_1, \dots, X_N \leq x_N)$$

(Ref: Page 23)

1.4 Joint PDF (Continuous Variables)

$$f_{X_1, \dots, X_N}(x_1, \dots, x_N) = \frac{\partial^N}{\partial x_1 \cdots \partial x_N} F_{X_1, \dots, X_N}(x_1, \dots, x_N)$$

(Ref: Page 24)

1.5 Independence Condition

Let:

$$\mathbf{X} = [X_1, \dots, X_N]^T$$

If independent:

$$f_X(x_1, \dots, x_N) = \prod_{i=1}^N f_{X_i}(x_i)$$

(Ref: Page 25)

2. JOINT GAUSSIAN RANDOM VARIABLES

2.1 Statement of Result (Joint PDF)

For two variables X, Y :

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(\frac{-1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) \right]\right)$$

(Ref: Page 3, 5)

2.2 Parameters

- Means: μ_X, μ_Y
- Standard deviations: σ_X, σ_Y
- Correlation coefficient: ρ

3. EXAMPLE 1 — JOINT GAUSSIAN

3.1 Given

- $\mu_X = 2, \sigma_X^2 = 1$
- $\mu_Y = -3, \sigma_Y^2 = 4$
- $\text{cov}(X, Y) = -1$

(Ref: Page 4)

3.2 (a) Joint PDF — Stepwise Derivation

Step 1: Convert Variances to Standard Deviations

$$\sigma_X = \sqrt{1} = 1, \quad \sigma_Y = \sqrt{4} = 2$$

(Ref: Page 8)

Step 2: Compute Correlation Coefficient

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{-1}{(1)(2)} = -\frac{1}{2}$$

(Ref: Page 7–8)

Step 3: Substitute into Joint Gaussian Formula

Values:

$$\mu_X = 2, \sigma_X = 1, \mu_Y = -3, \sigma_Y = 2, \rho = -0.5$$

(Ref: Page 9)

Step 4: Final Joint PDF Expression

$$f_{XY}(x, y) = \frac{1}{2\pi\sqrt{3}} \exp\left(-\frac{2}{3}\left[(x-2)^2 + \frac{(x-2)(y+3)}{2} + \frac{(y+3)^2}{4}\right]\right)$$

(Ref: Page 10)

3.3 (b) Marginal PDFs

Step 1: Integration Definition

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$$
$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$$

(Ref: Page 11)

Step 2: Result (Key Property)

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

(Ref: Page 11)

Step 3: Final Marginals

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-2)^2}{2}}$$
$$f_Y(y) = \frac{1}{2\sqrt{2\pi}} e^{-\frac{(y+3)^2}{8}}$$

(Ref: Page 12)

3.4 (c) Probability $P(X < 0)$

Step 1: Express Using CDF

$$P(X < 0) = F_X(0)$$

(Ref: Page 13)

Step 2: Standardization

$$Z = \frac{X - \mu_X}{\sigma_X}$$
$$P(X < 0) = P\left(Z < \frac{0 - 2}{1}\right) = P(Z < -2)$$

(Ref: Page 14)

Step 3: Symmetry Property

$$\Phi(-x) = 1 - \Phi(x) = P(Z > x)$$
$$P(Z < -2) = P(Z > 2)$$

(Ref: Page 15)

Step 4: Q-function Relation

$$Q(x) = P(Z > x)$$

Final Answer

$$P(X < 0) = Q(2)$$

(Ref: Page 16)

4. APPLICATION — LARGE MIMO SYSTEM

4.1 System Assumptions

- M transmit antennas
- N receive antennas

(Ref: Page 18–19)

4.2 Channel Model

Each path:

h_{ij} is a random variable

(Ref: Page 20)

4.3 Channel Matrix Representation

$$H = \begin{bmatrix} h_{11} & \dots & h_{1N} \\ h_{21} & \dots & h_{2N} \\ \vdots & \ddots & \vdots \\ h_{M1} & \dots & h_{MN} \end{bmatrix}$$

(Ref: Page 21)

5. EXAMPLE 2 — UNIFORM 3D VECTOR

5.1 Assumptions

$$|x_1| \leq 1, \quad |x_2| \leq 1, \quad |x_3| \leq 1$$

$$f_{X_1, X_2, X_3} = \begin{cases} C & \text{inside cube} \\ 0 & \text{otherwise} \end{cases}$$

(Ref: Page 26)

5.2 (a) Constant C

Step 1: Total Probability Condition

$$\int_{-1}^1 \int_{-1}^1 \int_{-1}^1 C \, dx_1 dx_2 dx_3 = 1$$

Step 2: Evaluate Integral

$$C(2)(2)(2) = 1$$

Step 3: Solve

$$C = \frac{1}{8}$$

(Ref: Page 28)

5.3 (b) Marginal f_{X_1, X_2}

$$f_{X_1, X_2} = \int_{-1}^1 \frac{1}{8} dx_3 = \frac{1}{8}(2) = \frac{1}{4}$$

(Ref: Page 29)

5.4 (c) Marginal f_{X_1}

$$f_{X_1} = \int_{-1}^1 \int_{-1}^1 \frac{1}{8} dx_2 dx_3$$

Step-by-step:

$$= \int_{-1}^1 \left(\frac{1}{8} \cdot 2 \right) dx_2 = \int_{-1}^1 \frac{1}{4} dx_2 = \frac{1}{4}(2) = \frac{1}{2}$$

(Ref: Page 31)

5.5 (d) Conditional PDF

$$f_{X_1|X_2,X_3} = \frac{f_{X_1,X_2,X_3}}{f_{X_2,X_3}} = \frac{1/8}{1/4} = \frac{1}{2}$$

(Ref: Page 33)

5.6 (e) Conditional PDF

$$f_{X_1,X_2|X_3} = \frac{f_{X_1,X_2,X_3}}{f_{X_3}} = \frac{1/8}{1/2} = \frac{1}{4}$$

(Ref: Page 35)

5.7 (f) Independence Proof

Condition

$$f_{X_1,X_2,X_3} = f_{X_1} f_{X_2} f_{X_3}$$

Step 1: LHS

$$\frac{1}{8}$$

Step 2: RHS

$$\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

Conclusion

Independent

(Ref: Page 37–39)

6. EXAMPLE 3 — TRIVARIATE DISTRIBUTION

6.1 Given

$$f_{XYZ}(x,y,z) = \begin{cases} Ke^{-(ax+by+cz)} & x,y,z > 0 \\ 0 & \text{otherwise} \end{cases}$$

(Ref: Page 40)

6.2 (a) Find K

Step 1: Total Probability

$$\int_0^\infty \int_0^\infty \int_0^\infty K e^{-(ax+by+cz)} dx dy dz = 1$$

Step 2: Separate Integrals

$$K \left(\int_0^\infty e^{-ax} dx \right) \left(\int_0^\infty e^{-by} dy \right) \left(\int_0^\infty e^{-cz} dz \right) = 1$$

Step 3: Evaluate

$$= \frac{1}{a} \cdot \frac{1}{b} \cdot \frac{1}{c}$$

Step 4: Solve

$$K = abc$$

(Ref: Page 42)

6.3 (b) Marginal f_{XY}

$$f_{XY}(x, y) = \int_0^\infty abce^{-(ax+by+cz)} dz$$

Step-by-step:

$$= abe^{-(ax+by)} \int_0^\infty ce^{-cz} dz = abe^{-(ax+by)}$$

(Ref: Page 44)

6.4 (c) Marginal f_X

$$\begin{aligned} f_X(x) &= \int_0^\infty abe^{-(ax+by)} dy \\ &= ae^{-ax} \end{aligned}$$

(Ref: Page 46)

6.5 (d) Independence Proof

Step 1: LHS

$$abce^{-(ax+by+cz)}$$

Step 2: RHS

$$(ae^{-ax})(be^{-by})(ce^{-cz}) = abce^{-(ax+by+cz)}$$

Conclusion

Independent

(Ref: Page 48–50)

END OF SCRIBE

This version:

- Preserves exact structure of slides
- Separates assumptions, results, and derivations clearly
- Expands every algebraic and integration step explicitly
- Uses only slide-provided content without any external additions